Personal Learning Roadmap – Next 90 Days (Post-Internship Career Acceleration)

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Baseline Completed (Day 30): Python, FastAPI, Pydantic v2, PyTest, RAG Pipelines (TForce India & CampusLLM), DevSecOps (LEGITIFY), Classical & Deep Learning (Solar Forecasting), and Hybrid Rules + 7-D Explainable ML (FinDoc-AuditEngine v3.0 at Linkific).
90-Day Objective: Transition from strong single-node AI/ML & Backend Microservices to Distributed Production AI Systems, Cloud-Native MLOps (Kubernetes/Kafka/vLLM), and High-Scale System Design ready for top-tier SDE / AI Engineer roles (Visa, Stripe, Enterprise Cloud & FinTech).

Roadmap Architecture Overview

flowchart LR
M1["Month 1 (Days 1-30)
Stateful Multi-Agent AI,
Hybrid RAG & Evaluation"] --> M2["Month 2 (Days 31-60)
Cloud-Native MLOps, K8s,
vLLM & Drift Telemetry"]
M2 --> M3["Month 3 (Days 61-90)
Distributed Event-Driven FinTech,
AppSec AI & System Design"]

Phase 1: Days 1–30 (Month 1) – Stateful Agentic Architectures, Hybrid RAG & Rigorous LLM Evaluation

Week	Days	Core Technical Theme	Actionable Study & Implementation Tasks	Concrete Deliverable / KPI
Week 1	Days 1–7	Hybrid Retrieval (BM25 + Dense Vector + Cross-Encoder Reranking)	Implement Reciprocal Rank Fusion (RRF) combining sparse BM25 keyword matching (for exact SKU/PO codes) with dense vector embeddings (bge-large-en / Qdrant) and ms-marco cross-encoder reranking.	Upgrade CampusLLM & FinDoc-AuditEngine policy search with RRF; benchmark Recall@5 improvement (>= +18%).
Week 2	Days 8–14	Stateful Multi-Agent Workflows (LangGraph & Tool Calling)	Build cyclic, state-persisted agent graphs with deterministic checkpointing, human-in-the-loop (HITL) approval interrupts, and strict JSON Schema tool definitions.	Autonomous Procurement Dispute Resolution Agent that drafts vendor chargeback memos from FinDoc-AuditEngine item diffs.
Week 3	Days 15–21	Automated AI Evaluation (RAGAS, DeepEval & Golden Datasets)	Build CI-integrated evaluation harnesses measuring Faithfulness, Context Precision, Answer Relevance, and Hallucination Rate across 100+ adversarial test prompts.	GitHub Actions quality gate that fails PRs if faithfulness drops below 0.92 or latency exceeds P95 budget.
Week 4	Days 22–30	Multimodal Document AI (LayoutLMv3 / Vision-Language OCR)	Replace regex-only text extraction with layout-aware bounding-box table extraction for scanned multi-page PDF GST/VAT invoices and bills of lading.	End-to-end PDF-to-InvoicePayload extraction microservice with token-level confidence calibration.

Phase 2: Days 31–60 (Month 2) – Cloud-Native MLOps, High-Throughput Inference & Drift Observability

Week	Days	Core Technical Theme	Actionable Study & Implementation Tasks	Concrete Deliverable / KPI
Week 5	Days 31–37	Async Persistence (PostgreSQL + SQLAlchemy 2.0 + asyncpg + Redis)	Migrate in-memory ledgers to partitioned PostgreSQL tables with Alembic migrations, optimistic concurrency locking, and Redis idempotency keys (24h TTL).	Zero-duplicate transactional audit ledger supporting 1,000+ concurrent req/sec under locust load testing.
Week 6	Days 38–44	Model Registry, Versioning & Drift Monitoring (MLflow + EvidentlyAI)	Version StandardScaler + IsolationForest artifacts in MLflow; compute weekly Population Stability Index (PSI) and Wasserstein Distance across the 7 financial features.	Automated drift alert webhook triggered whenever $\text{PSI} > 0.20$ on vendor price variance or tax distributions.
Week 7	Days 45–52	High-Throughput Model Serving (ONNX Runtime & vLLM)	Export scikit-learn/XGBoost pipelines to ONNX Runtime for sub-millisecond C++ execution; benchmark quantized LLM serving (AWQ/GPTQ) with continuous batching.	Reduce 7-D anomaly + perturbation XAI inference latency from 7.9 ms to < 1.8 ms via vectorized batch perturbation.
Week 8	Days 53–60	Kubernetes (EKS/AKS), Docker Multi-Stage Builds & OpenTelemetry	Package services into distroless multi-stage Docker images (<150 MB), write Helm charts with Horizontal Pod Autoscalers (HPA), and instrument Prometheus + Grafana dashboards.	Production K8s deployment manifest with P50/P95/P99 latency histograms and automated liveness/readiness probes.

Phase 3: Days 61–90 (Month 3) – Event-Driven FinTech Scale, AppSec AI & Career Execution

Week	Days	Core Technical Theme	Actionable Study & Implementation Tasks	Concrete Deliverable / KPI
Week 9	Days 61–67	Event-Driven Streaming (Apache Kafka / RabbitMQ + Celery)	Decouple high-volume ERP invoice ingestion from downstream ML scoring and webhook notifications using durable Kafka topics and dead-letter queues (DLQ).	Event-driven FinDoc-AuditEngine v4.0 handling burst loads of 10,000 invoices/min with zero message loss.
Week 10	Days 68–75	AI Security, Adversarial Robustness & OWASP LLM Top 10	Combine LEGITIFY (AppSec) with

FinDoc-AuditEngine: defend against prompt injection inside invoice descriptions, data poisoning, and model inversion attacks. | Security hardening suite & automated red-team adversarial test harness (50+ injection payloads blocked). |
| Week 11 | Days 76–82 | Distributed System Design & Algorithmic Mastery | Solve 60+ medium/hard DSA problems (Graphs, Dynamic Programming, Heaps, Tries) and author 6 end-to-end system design architectures (Payment Switch, Real-Time Fraud Engine, Distributed Rate Limiter). | Complete System Design & Coding Interview Portfolio repository with benchmarked implementations. |
| Week 12 | Days 83–90 | Open-Source Contribution, Technical Writing & Panel Readiness | Publish 2 deep-dive engineering articles (*"Perturbation XAI for Unsupervised Financial Anomaly Detection"* and *"Pre-Inference PII Gateways in FastAPI"*) and contribute a PR to an open-source Python/AI library. | Published engineering blog posts, merged OSS pull request, and full interview readiness for tier-1 SDE/AI roles. |

90-Day Success Metrics Summary

1. System Performance: Reduce FinDoc-AuditEngine P99 inference latency to < 3 ms via vectorized ONNX batching and scale throughput to 1,000+ RPS backed by PostgreSQL + Redis.
2. Engineering Rigor: Maintain >= 95% test coverage across all portfolio repositories with automated CI/CD quality and security gates.
3. Career Readiness: Complete 60+ targeted DSA problems, 6 FinTech/AI system design blueprints, and 2 published technical articles showcased on LinkedIn and GitHub.